

Learning Path Recommendation System

Personalized Course Recommendations using Collaborative Filtering and Matrix Factorization (TruncatedSVD)

Project Report

Submitted by: **Alishba**

1. Introduction

The Learning Path Recommendation System is a Machine Learning-based system that gives personalized course recommendations to interns. It studies earlier intern-course learning interactions and suggests courses that an intern is likely to be interested in.

The project uses **Collaborative Filtering** with **Matrix Factorization**. Truncated Singular Value Decomposition (**TruncatedSVD**) finds hidden patterns in the intern-course data and produces predicted scores for courses that an intern has not yet taken.

2. Project Objective

- Analyze intern learning interactions.
- Create an intern-course interaction matrix.
- Identify hidden learning patterns using Matrix Factorization.
- Predict potential interest in unseen courses.
- Generate personalized Top-5 course recommendations for each intern.
- Show that different interns receive different recommendations.

3. Dataset

The project uses a simulated learning interaction dataset with the following size:

Item	Count
Interns	200
Courses	30
Recorded interactions	2,808

The dataset holds details about interns, courses, course categories, difficulty levels, duration, and interaction scores.

Interaction Matrix

- **Rows** = Interns
- **Columns** = Courses
- **Values** = Interaction scores

Scores range from **1 to 5**; a higher value means a stronger learning interaction. A value of **0** means **no recorded interaction**, i.e. the intern has not yet taken or interacted with that course.

4. Methodology

Dataset → Interaction Matrix → Matrix Factorization → Predicted Scores → Course Ranking → Personalized Recommendations

4.1 Collaborative Filtering

Collaborative Filtering recommends items by using existing user-item interaction patterns. Here, users are interns, items are courses, and interactions are learning interaction scores. The system learns from courses an intern has already used and estimates their interest in other courses.

4.2 Matrix Factorization

Matrix Factorization describes the large intern-course matrix using a smaller number of hidden (**latent**) factors. These factors capture learning patterns that are not directly visible in the raw data. For example, the model may learn that some interns show interaction patterns related to AI/ML, Programming, or Generative AI.

4.3 TruncatedSVD

TruncatedSVD performs the Matrix Factorization, configured with **10 latent components**. It reduces the interaction matrix to a lower-dimensional form, which is then used to rebuild estimated interaction scores. These predicted scores rank the courses and select the Top-5 for each intern.

5. Recommendation Generation

After training, predicted scores were generated for the intern-course combinations. For each intern, courses were ranked by predicted score and the **Top 5** were chosen as recommendations. With 200 interns and 5 recommendations each, the final dataset has **$200 \times 5 = 1,000$ recommendations**.

Each result contains:

- Intern_ID
- Course_ID
- Course_Name
- Category
- Difficulty
- Duration_Hours
- Predicted_Score

6. Model Evaluation

The model was evaluated by comparing predicted interaction scores with the observed values.

Metric	Result
MAE	1.0088
MSE	1.5514
RMSE	1.2455

Interpretation

MAE (Mean Absolute Error) is about **1.01**: predicted scores differ from actual scores by roughly 1 point on average. **RMSE** is about **1.25** and gives more weight to larger errors. Together they show how closely the predicted scores match the observed interactions.

7. Personalization Demonstration

Recommendations were generated for five different interns to show personalization.

Intern	Top Recommended Course	Predicted Score
INT001	Technical Interview Preparation	2.404
INT002	SQL Fundamentals	2.060
INT003	AWS Fundamentals	2.310
INT004	Git and GitHub	2.503
INT005	Feature Engineering	2.307

The top courses differ between interns. For example, INT001 received **Technical Interview Preparation**, INT004 received **Git and GitHub**, and INT005 received **Feature Engineering**. The system does not give every intern the same list; recommendations come from the learned interaction patterns of each intern.

8. Final Output

The final file is **all_intern_recommendations.csv**. It holds **1,000 Top-5 recommendations** for the 200 interns, with the predicted score and course details such as category, difficulty, and duration.

9. Limitations

- The dataset is simulated, not collected from a real learning platform.
- The number of interns and interactions is fairly small.
- A score of zero means no recorded interaction; it does not mean the intern dislikes the course.
- New interns or courses with little or no history are hard to recommend accurately.
- The system uses mainly interaction patterns, not personal learning preferences or detailed course content.

10. Future Improvements

- Use a larger, real-world learning dataset.
- Include course content and descriptions.
- Combine collaborative filtering with content-based filtering.
- Add explicit intern preferences and career goals.
- Try more advanced recommendation algorithms.
- Update recommendations as new interactions are recorded.
- Evaluate with more metrics such as Precision@K and Recall@K.

11. Conclusion

The Learning Path Recommendation System shows how Machine Learning can give personalized learning recommendations. Using **Collaborative Filtering with Matrix Factorization** (TruncatedSVD), it learns hidden patterns from an intern-course interaction matrix, estimates interest in unseen courses, and produces personalized Top-5 recommendations. The results show different recommendations for different interns, which confirms the personalization. The project is a practical example of how recommendation systems can support personalized learning paths for interns.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- TruncatedSVD
- Matrix Factorization
- Collaborative Filtering
- CSV Dataset
- Machine Learning

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